PICKLEBALL ROBOT CONTROLLER — EIGHT READABLE KICAD SHEETS

00 Battery Safety

File: 00_battery_safety.kicad_sch

01 Power and Pi

File: 01_power_and_pi.kicad_sch

02 ESP32 and Pi UART

File: 02_esp32_and_pi.kicad_sch

03 Left Motor

File: 03_left_motor.kicad_sch

04 Right Motor

File: 04_right_motor.kicad_sch

05 Encoders

File: 05_encoder_inputs.kicad_sch

06 Arm Servos

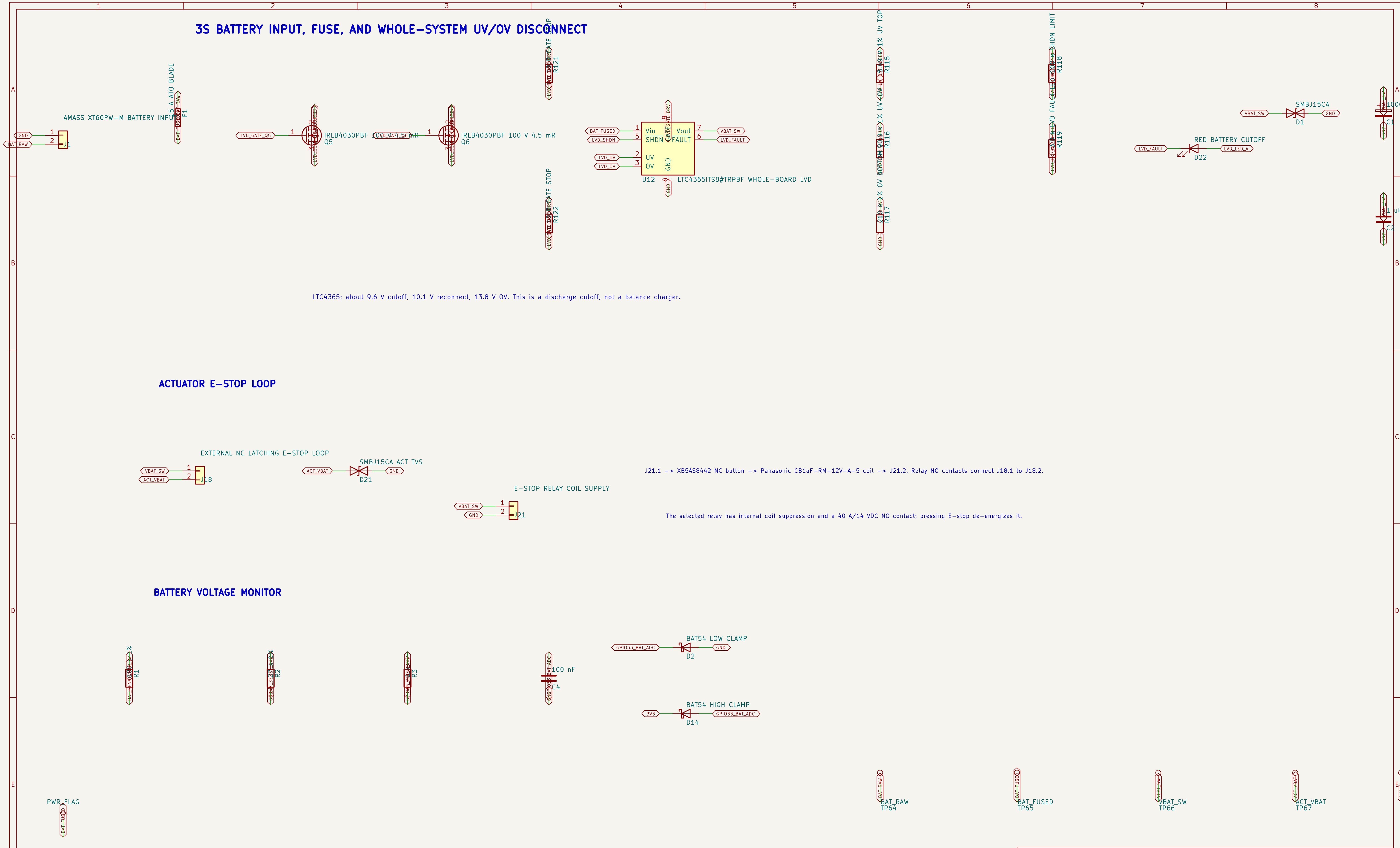
File: 06_arm_servos.kicad_sch

07 Main Servo 8.4V

File: 07_main_servo_8v4.kicad_sch

Global net labels connect the sheets. Open any sheet from the hierarchy panel.

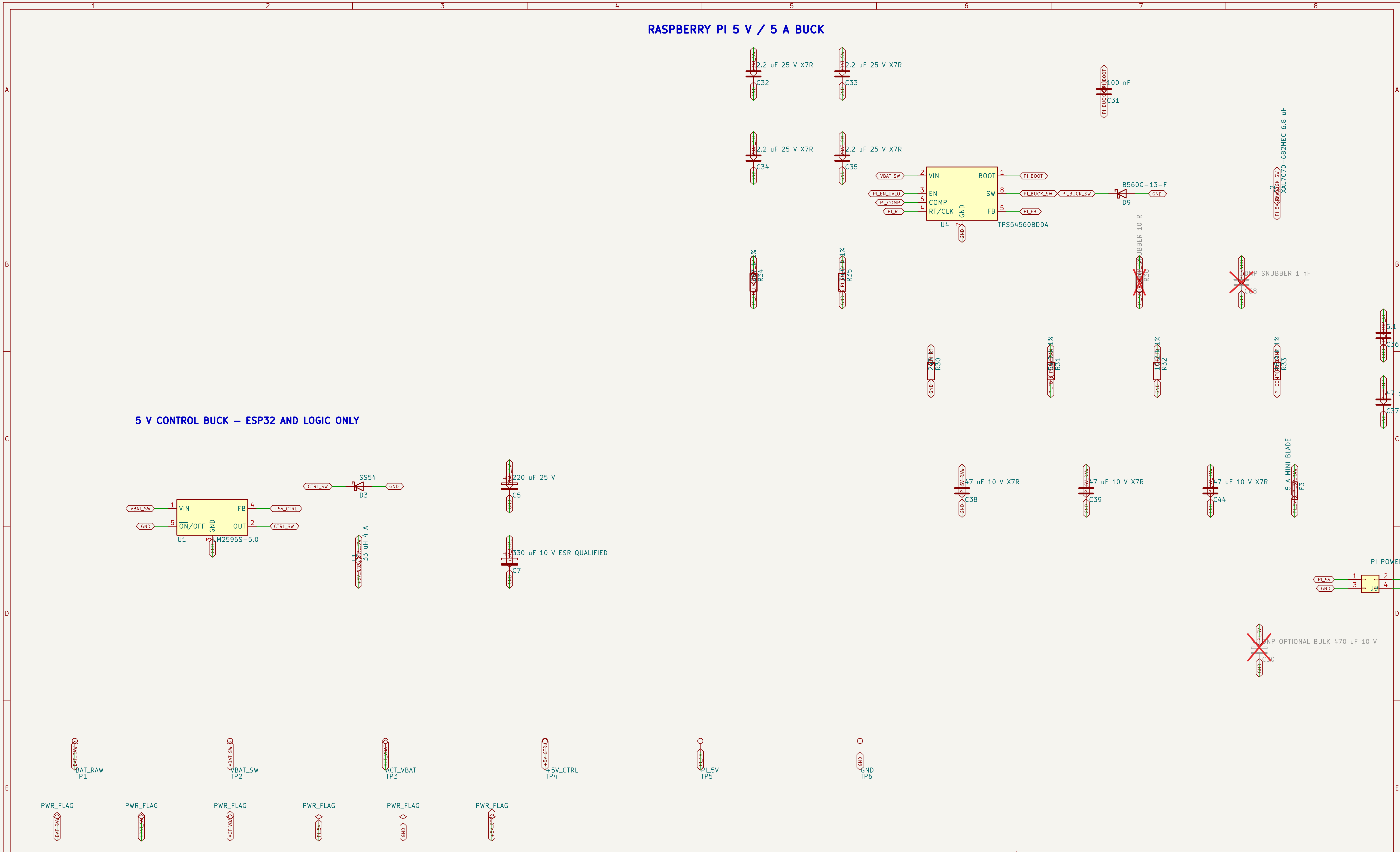
Harrison		
Sheet: / File: pickleball_robot_controller.kicad_sch		
Title: Pickleball Robot Controller — Hierarchical Overview		
Size: A4	Date: 2026-07-12	Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4		Id: 1/9



Sheet 1 of 8
Harrison
Sheet: /00 Battery Safety/
File: 00_battery_safety.kicad_sch

Title: Battery Fuse, Whole–System Cutoff, and E–Stop

Size: A3 Date: 2026-07-12 **Rev: R2–MULTISHEET**
KiCad E.D.A. 10.0.4 Id: 2/9



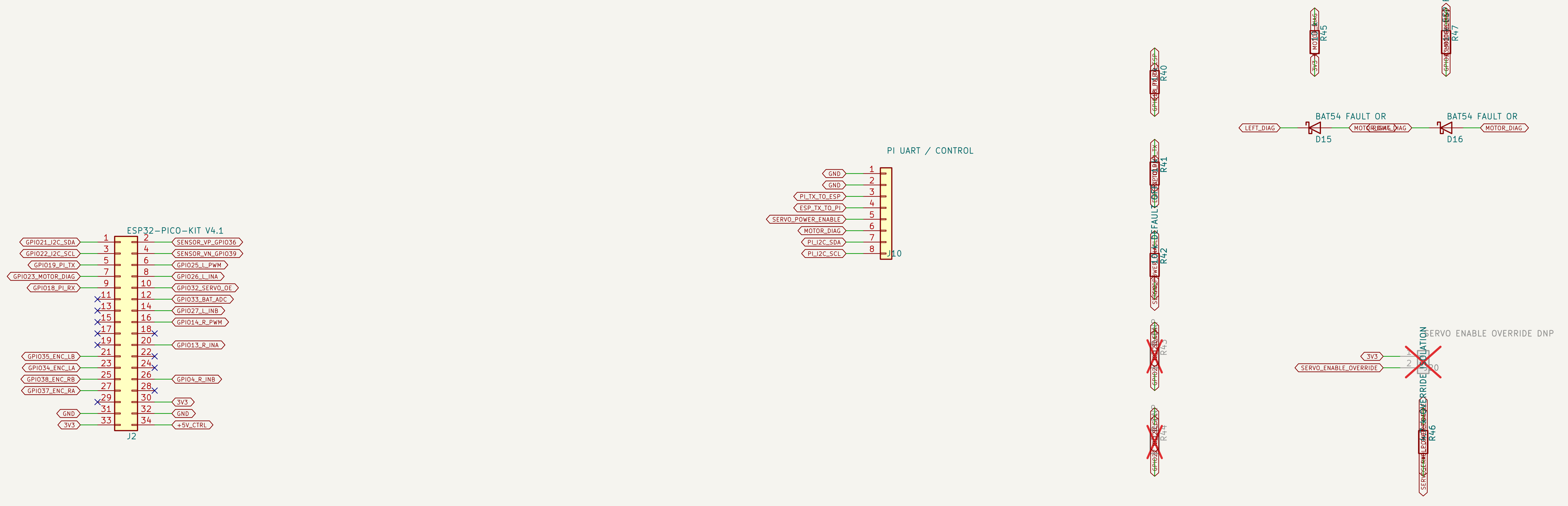
Sheet 2 of 8
Harrison
Sheet: /01 Power and Pi/
File: 01_power_and_pi.kicad_sch

Title: Logic Buck and Raspberry Pi Power

Size: A3 | Date: 2026-07-12 | Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4 | Id: 3/9

ESP32-PICO-KIT V4.1 CARRIER — CONFIRMED 17.78 mm ROW SPACING

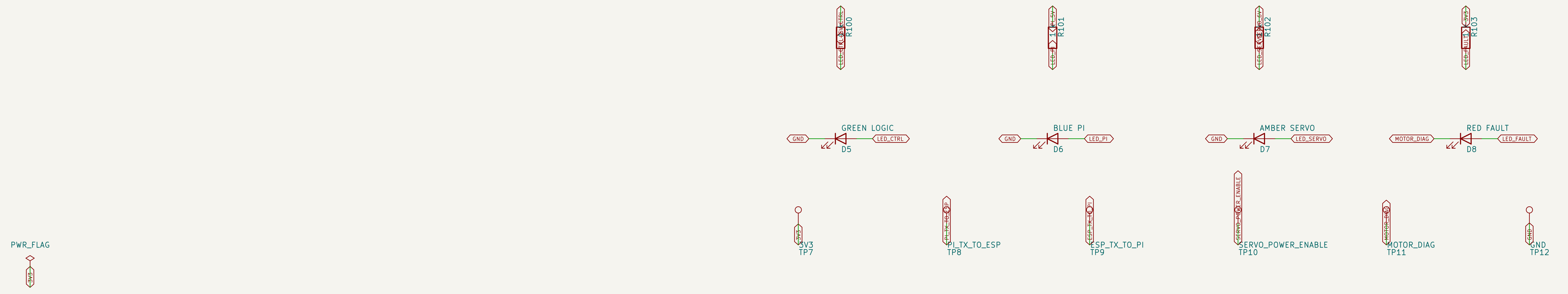
RASPBERRY PI UART / CONTROL — NORMAL 2.54 mm HEADER



J10: 1/2 GND; 3 Pi TX->ESP GPIO18; 4 ESP GPIO19->Pi RX; 5 servo-power enable; 6 motor fault; 7/8 I2C DNP.

Pi: GPIO14 pin8 -> J10.3; GPIO15 pin10 <- J10.4; GPIO17 pin11 -> J10.5; GPIO27 pin13 <- J10.6; GND -> J10.1/2.

STATUS LEDS

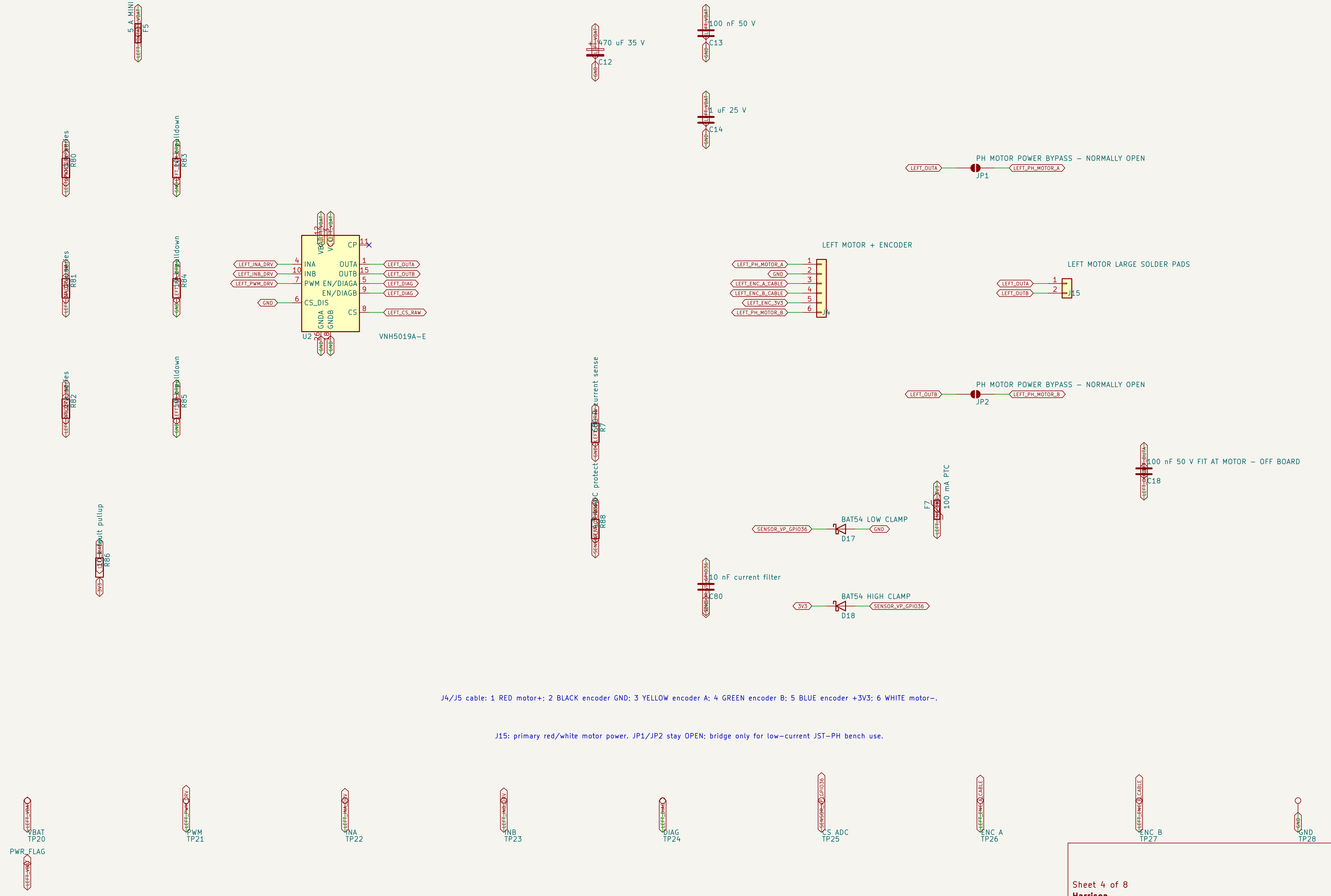


Sheet 3 of 8
Harrison
Sheet: /02 ESP32 and Pi UART/
File: 02_esp32_and_pi.kicad_sch

Title: ESP32 Carrier, Raspberry Pi UART, and Status LEDs

Size: A3	Date: 2026-07-12	**Rev: R2-MULTISHEET**
KiCad E.D.A. 10.0.4		Id: 4/9

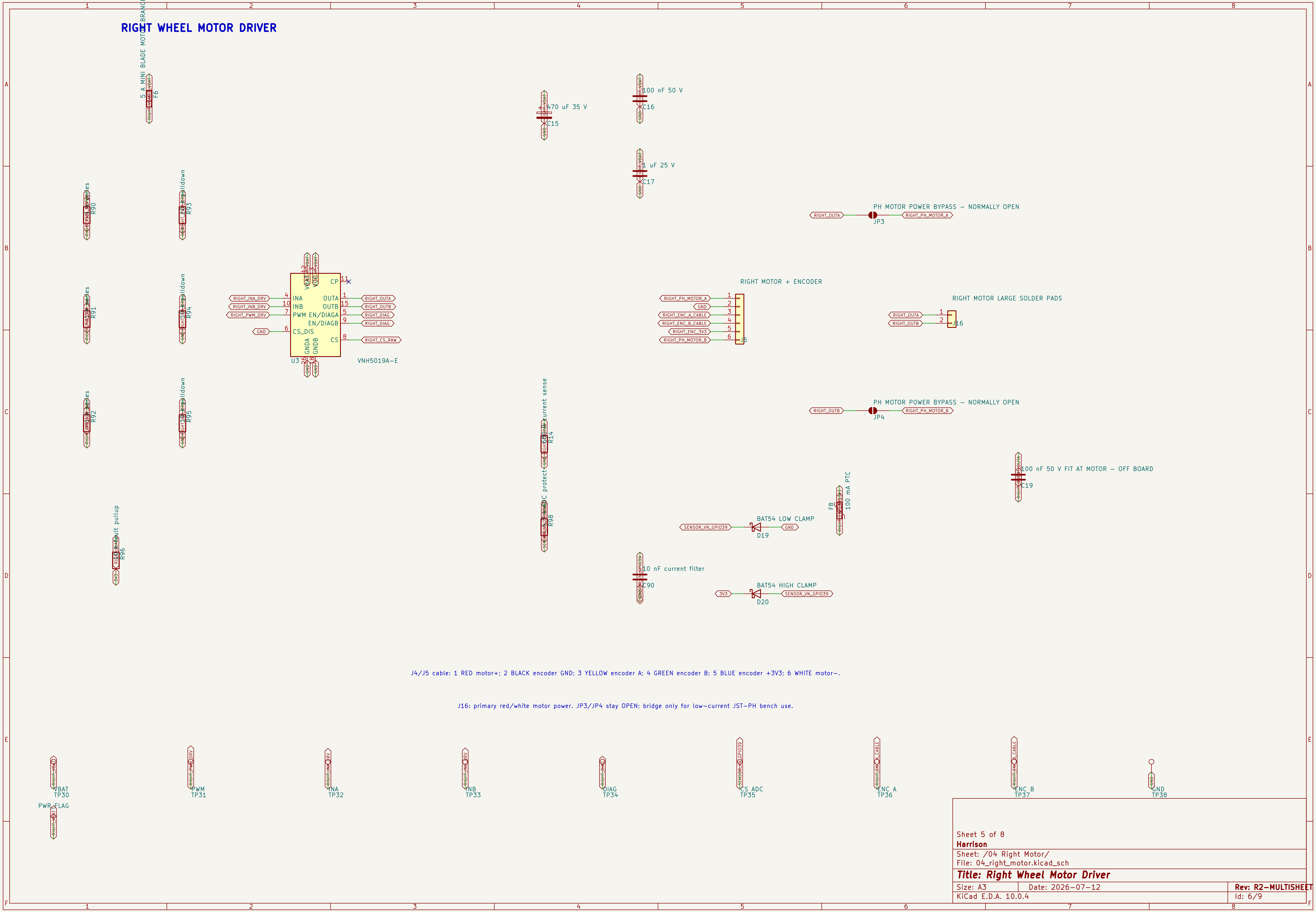
LEFT WHEEL MOTOR DRIVER



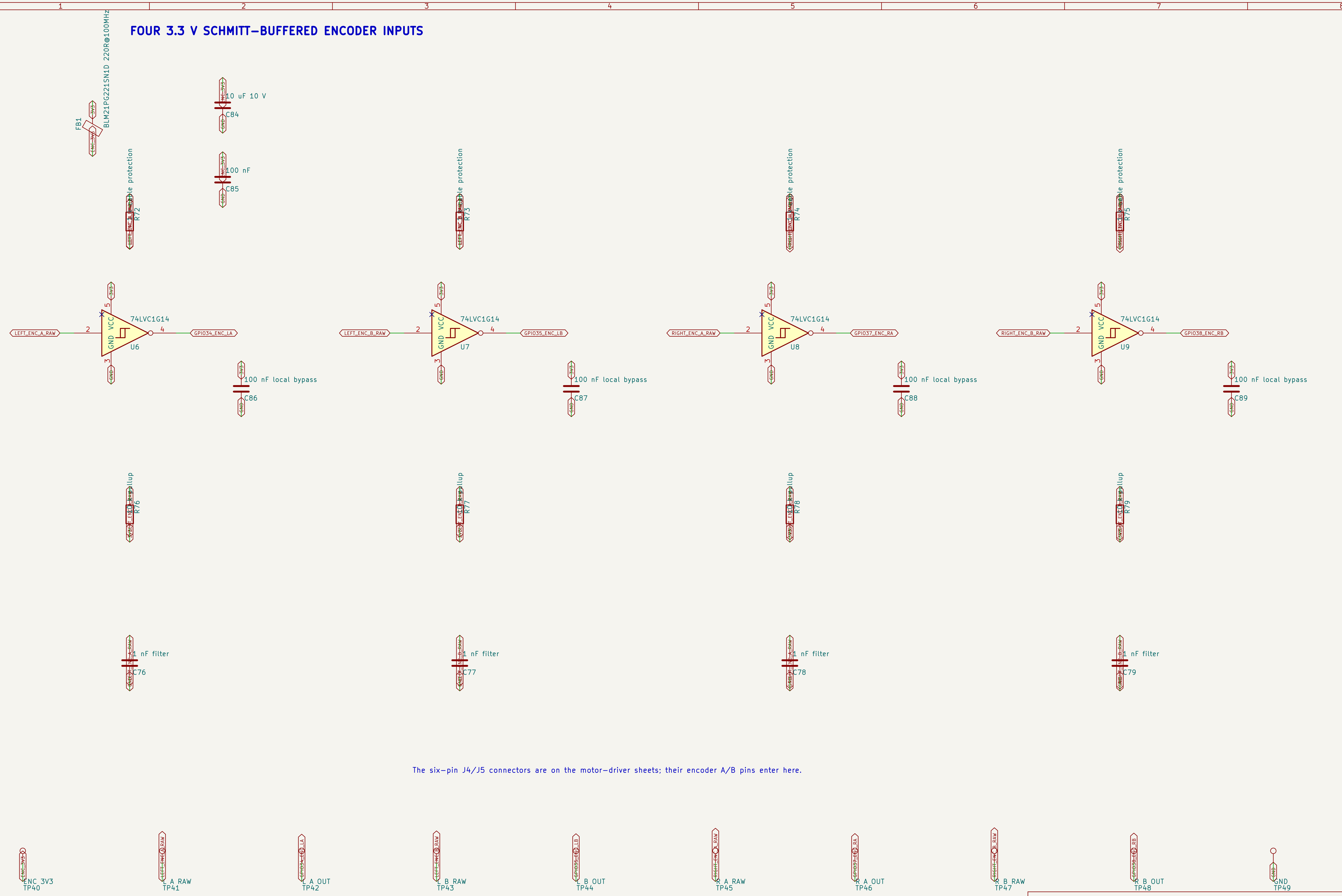
J4/J5 cable: 1 RED motor+; 2 BLACK encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor–.

J15: primary red/white motor power. JP1/JP2 stay OPEN; bridge only for low–current JST–PH bench use.

Sheet 4 of 8
Harrison
Sheet: /03 Left Motor/
File: 03_left_motor.kicad_sch
Title: Left Wheel Motor Driver
Size: A3 Date: 2026-07-12 Rev: R2–MULTISHEET
KiCad E.D.A. 10.0.4 Id: 5/9


RIGHT WHEEL MOTOR DRIVER
5 A MINI BLADE MOTOR BRANCH
F6
470 uF 35 V
C15
100 nF 50 V
C16
1 uF 25 V
C17
R90
R93
R91
R94
R92
R95
R96
R14
R98
INA
INB
PWM
CS_DIS
CP
OUTA
OUTB
EN/DIAGA
EN/DIAGB
CS
GNDA
GNDB
U3
VNH5019A-E
RIGHT_INA_DRV
RIGHT_INB_DRV
RIGHT_PWM_DRV
GND
RIGHT_OUTA
RIGHT_OUTB
RIGHT_DIAG
RIGHT_CS_RAW
PH MOTOR POWER BYPASS - NORMALLY OPEN
JP3
RIGHT_PH_MOTOR_A
RIGHT MOTOR + ENCODER
RIGHT_ENC_A_CABLE
RIGHT_ENC_B_CABLE
RIGHT_ENC_3V3
RIGHT_PH_MOTOR_B
J5
RIGHT MOTOR LARGE SOLDER PADS
J16
PH MOTOR POWER BYPASS - NORMALLY OPEN
JP4
100 nF 50 V FIT AT MOTOR - OFF BOARD
C19
100 mA PTC
F8
BAT54 LOW CLAMP
SENSOR_VN_GPIO39
D19
10 nF current filter
C90
BAT54 HIGH CLAMP
3V3
D20
J4/J5 cable: 1 RED motor+; 2 BLACK encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor-.
J16: primary red/white motor power. JP3/JP4 stay OPEN; bridge only for low-current JST-PH bench use.
VBAT
TP30
PWR_FLAG
PWM
TP31
INA
TP32
INB
TP33
DIAG
TP34
CS_ADC
TP35
ENC_A
TP36
ENC_B
TP37
GND
TP38
Sheet 5 of 8
Harrison
Sheet: /04 Right Motor/
File: 04_right_motor.kicad_sch
Title: Right Wheel Motor Driver
Size: A3
Date: 2026-07-12
Rev: R2-MULTISHEET
KiCad E.D.A. 10.0.4
Id: 6/9

FOUR 3.3 V SCHMITT-BUFFERED ENCODER INPUTS



Sheet 6 of 8
Harrison
Sheet: /05 Encoders/
File: 05_encoder_inputs.kicad_sch

Title: Wheel Encoder Input Conditioning

Size: A3	Date: 2026-07-12	**Rev: R2-MULTISHEET**
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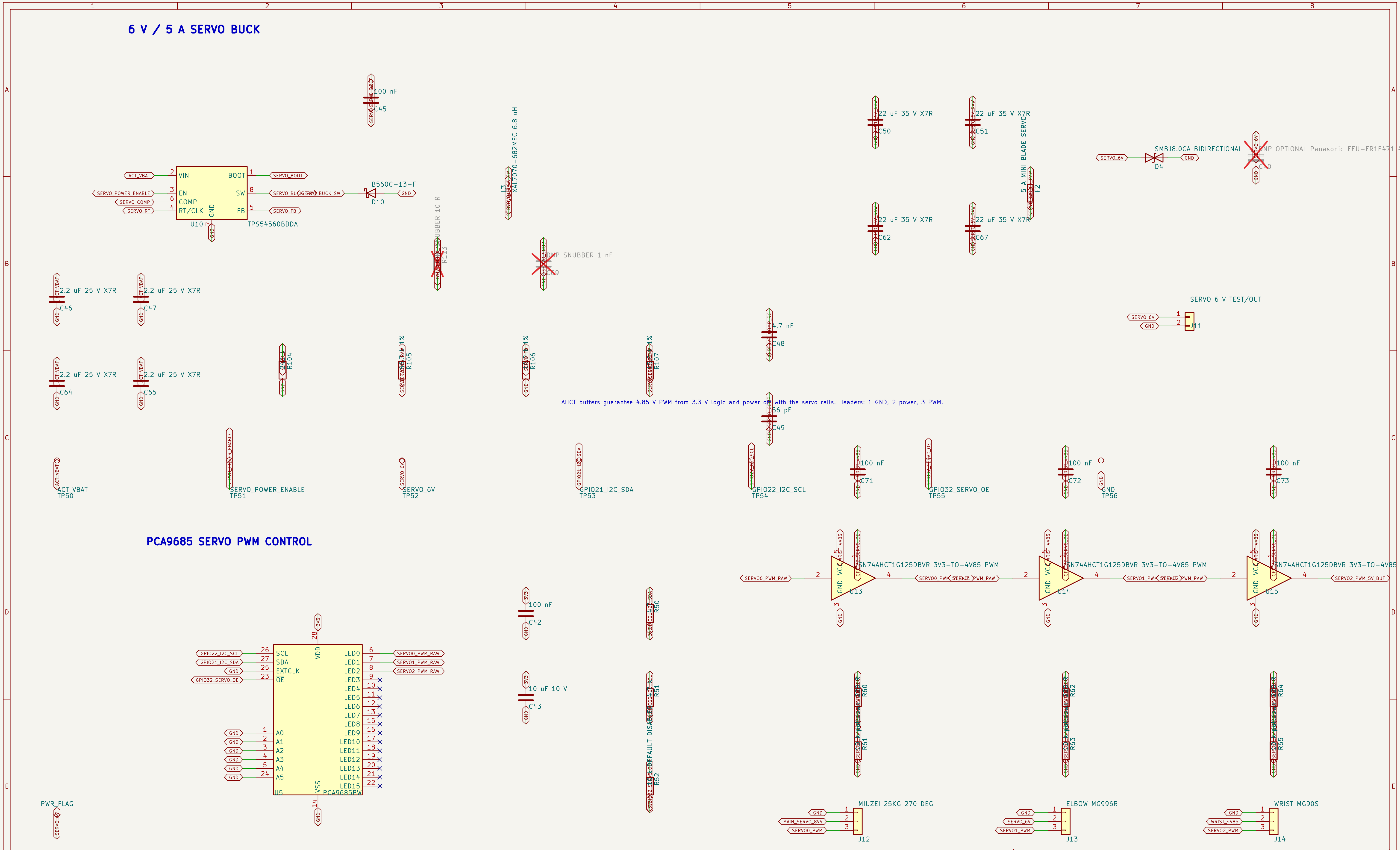

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Harrison

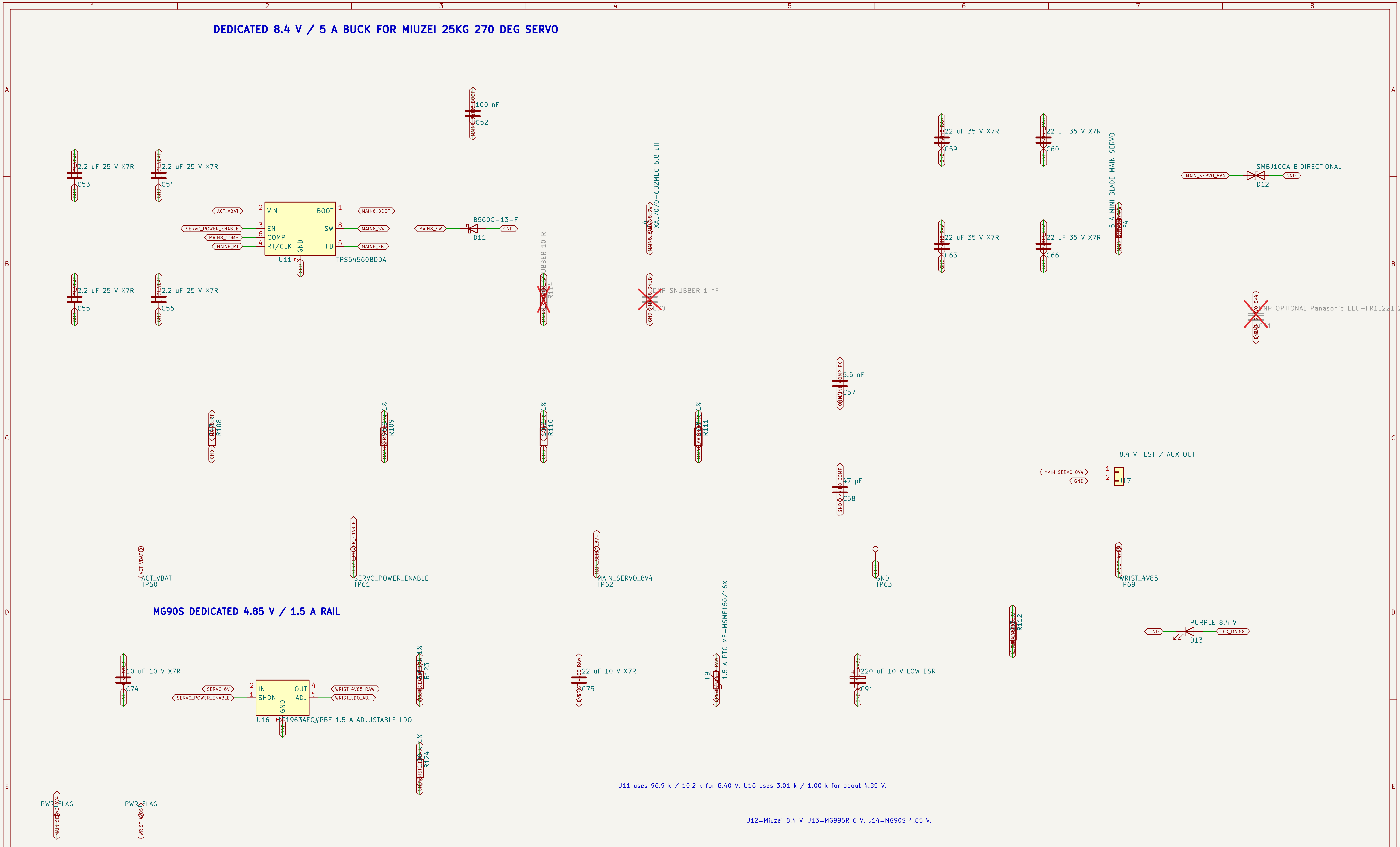
Sheet: /06 Arm Servos/

File: 06_arm_servos.kicad_sch

Title: Servo Power Regulator and 3-DOF Arm Control

Size: A3 Date: 2026-07-12 Rev: R2-MULTISHEET

KiCad E.D.A. 10.0.4 Id: 8/9



Sheet 8 of 8

Harrison

Sheet: /07 Main Servo 8.4V/

File: 07_main_servo_8v4.kicad_sch

Title: Miuzei 8.4 V and MG90S 4.85 V Power Rails

Size: A3 | Date: 2026-07-12 | Rev: R2-MULTISHEET

KiCad E.D.A. 10.0.4 | Id: 9/9